

Unit 1 Progress Check: FRQ

Name _____

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to **K**.

- (a) In terms of atomic structure, explain why the atomic radius of **K** is larger than that of **Na**.



Please respond on separate paper, following directions from your teacher.

- (b) In terms of atomic structure, explain why the first-ionization energy **K** is less than that of **Ca**.



Please respond on separate paper, following directions from your teacher.

- (c) **K** forms the compound **K₂O**, which is an ionic compound that is brittle. Identify another element, **M**, that is likely to form a brittle, ionic compound with the formula **M₂O**. Justify your answer in terms of periodic trends.



Please respond on separate paper, following directions from your teacher.

- (d) The compound **K₂O₂** also exists. A chemist can determine the mass of **K** in a sample of known mass that consists of either pure **K₂O** or pure **K₂O₂**. From this information, can the chemist answer the question of which compound is in the sample? Indicate yes or no, and explain.



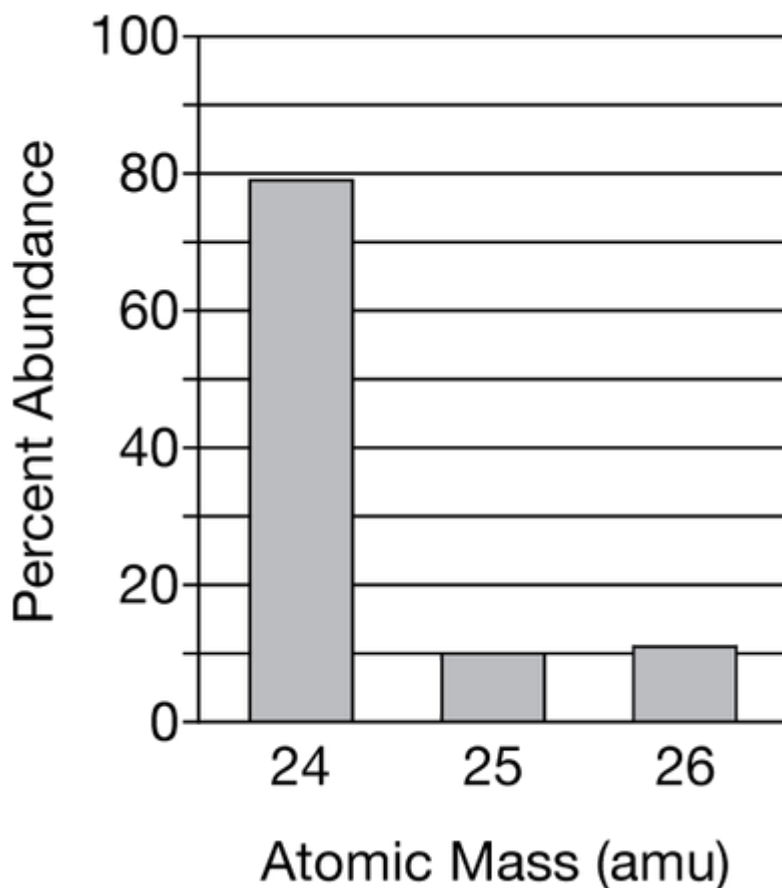
Please respond on separate paper, following directions from your teacher.

2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



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A sample of a pure element is analyzed using a mass spectrometer. The results are shown below.



(a) How many different isotopes of the element were in the sample?



Please respond on separate paper, following directions from your teacher.

(b) Describe how to use the information from the mass spectrum to determine the average atomic mass of the element.



Please respond on separate paper, following directions from your teacher.

(c) Identify the element.



Please respond on separate paper, following directions from your teacher.



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- (d) Write the ground-state electron configuration of an atom of the element that you identified in part (c).



Please respond on separate paper, following directions from your teacher.